Mandatory deliverable

LAB | AWS VPC Peering Lab

Skills you will learn

Cloud Computing: Compute & Networking

Introduction

In this hands-on lab, you will establish a **VPC Peering Connection** between two VPCs across different AWS regions. This will help you understand how to enable communication between VPCs while maintaining security and control over network traffic.

Learning Objectives

By completing this lab, you will be able to:

1. **Create two VPCs** in different AWS regions.
2. **Set up a VPC Peering Connection** to enable cross-region communication.
3. **Modify Route Tables** to direct traffic between peered VPCs.
4. **Verify connectivity** between resources in the peered VPCs.

Lab Instructions

1. Create Two VPCs in Separate Regions

1. Navigate to **AWS Management Console** → **VPC Service**.
2. Select **US East (N. Virginia) (us-east-1)** region.
3. Click **Create VPC** and enter the following details:
   - **VPC Name**: `main-vpc`
   - **IPv4 CIDR Block**: `10.0.0.0/16`
   - Click **Create VPC**.
4. Switch to **US East (Ohio) (us-east-2)** region.
5. Click **Create VPC** and enter the following details:

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- **VPC Name**: `sub-vpc`
- **IPv4 CIDR Block**: `10.11.0.0/16`
- Click **Create VPC**.

2. Create a VPC Peering Connection

1. In **US East (N. Virginia) (us-east-1)**, navigate to **VPC Peering Connections**.
2. Click **Create Peering Connection** and enter the following details:
   - **Name**: `prod-peer`
   - **VPC ID (Requester)**: Select `main-vpc`
   - **Account**: Select **My Account**
   - **Region**: Choose **Another Region** → Select **US East (Ohio) (us-east-2)**.
   - **VPC ID (Accepter)**: Select `sub-vpc`
3. Click **Create Peering Connection**.

3. Accept the Peering Connection

1. Switch to **US East (Ohio) (us-east-2)** region.
2. Navigate to **VPC Peering Connections**.
3. Find the peering connection with the status `Pending Acceptance` .
4. Click **Accept Peering Request**.

4. Configure Route Tables

1. Navigate to **Route Tables** in both regions.
2. **For `main-vpc` in US East (N. Virginia) (us-east-1)**:
   - Select **Route Table** → **Edit Routes**.
   - Add a new route:
     - **Destination**: `10.11.0.0/16`
     - **Target**: Select the **Peering Connection (`prod-peer`)**
   - Click **Save Changes**.
3. **For `sub-vpc` in US East (Ohio) (us-east-2)**:
   - Select **Route Table** → **Edit Routes**.
   - Add a new route:
     - **Destination**: `10.0.0.0/16`
     - **Target**: Select the **Peering Connection (`prod-peer`)**
   - Click **Save Changes**.

5. Verify Connectivity

1. Launch an **EC2 instance** in `main-vpc` and another in `sub-vpc`.
2. Assign **private IP addresses** to both instances.
3. SSH into one instance and attempt to `ping` the private IP of the other instance:

```
ping <private-IP-of-other-instance>
```

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4. If connectivity fails, check the **security groups** and allow **ICMP traffic** between the VPCs.

Completion Criteria

- **Two VPCs created** in separate AWS regions.
- **VPC Peering Connection** successfully established.
- **Route Tables updated** to enable communication.
- **Successful connectivity test** between EC2 instances in different VPCs.

Submission Guidelines

- Take **screenshots** of key steps and results.
- Compile them into a **Google Doc** and upload them to **Google Drive**.
- Share the **Google Drive link** with the instructor and ensure they have access.

🚀 **Congratulations! You've successfully set up VPC Peering across AWS regions!** 🎉

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Lab

URL

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